Handle Request Methods

I need to have a tool to handle all of the request methods, like Postman or Insomnia

Structure of API

API Endpoint

It is the URL that represents specified resource or function in API

Return Response to Server

```
from flask import Flask, Response
import json
import os

app = Flask(__name__)


@app.route("/api/v1.0/studnets", methods = ["GET"])

def students():
    student_list = [
        {
            "name": "Jevon",
            "country": "Indonesia",
            "skills": ["Coding", "Workout", "Trading"]
        },
        {
            "name": "Alvaro",
            "country": "Indonesia",
            "skills": ["Trading", "Communicating"]
        }
    ]

    return Response(json.dumps(student_list), mimetype="application/json")
```

Return Response of Full Data from MongoDB

```
MONGODB_URL = ...
client = pymongo.MongoClient(MONGODB_URL)
db = client["thirty_days_of_python"]
...
def students():
    data = list(db.students.find())
```

```
    ## It is said that MongoDB _id is not serializable, cannot encode
ObjectId, so convert it to string first
    for item in data:
        item["_id"] = str(item["_id"])

    return Response(json.dumps(data), mimetype="application/json")
```

Return Response of Data with Specified Id

```
## Flask should accept anything in the url <id> and store it in a variable
called id
@app.route("/api/v1.0/students/<id>", methods = ['GET'])

def single_student(id):

    student = db.students.find_one({"_id": ObjectId(id)})

    if student:

        student["_id"] = str(student["_id"])

        return Response(json.dumps(student), mimetype="application/json")

    else:

        return Response(json.dumps({"error": "student not found"}),
status=404, mimetype="application/json")
```

Create Data by POST

```
@app.route("/api/v1.0/students", methods = ["POST"])

def create_student():

    name = request.form["name"]

    country = request.form["country"]

    city = request.form["city"]

    age = int(request.form["age"])

    student = {

        "name": name,
```

```
        "country": country,
        "city": city,
        "age": age
    }
    db.students.insert_one(student)
    return
```

Update Data by PUT

```
@app.route("/api/v1.0/students/<id>", methods = ["PUT"])
def update_student(id):
    query = {"_id": ObjectId(id)}
    name = request.form["name"]
    country = request.form["country"]
    city = request.form["city"]
    age = int(request.form["age"])
    student = {
        "name": name,
        "country": country,
        "city": city,
        "age": age
    }
    db.students.update_one(query, student)
    return Response(json.dumps({"result": "a student has been updated"}))
```

Delete Data by DELETE

```
@app.route("/api/v1.0/students/<id>", methods = ["DELETE"])

def delete_student(id):

    db.students.delete_one({"_id": ObjectId(id)})

    return Response(json.dumps({"result": "a student has been deleted"}))
```